Dynamic Soft-Thresholding for Feature Selection in Regression

by SRI KARTHIK MANNE

Submission date: 31-Mar-2026 12:36PM (UTC+0530)
Submission ID: 2918601633
File name: NOP.pdf (758.82K)
Word count: 1968
Character count: 11600

Dynamic Soft-Thresholding for Feature Selection in Regression

PROJECT REPORT

NAME : SRI KARTHIK MANNE

USN : 23BTRCL079

in partial fulfilment for the award of the degree of

BACHELOR OF ENGINEERING

in

B. Tech. Computer Science and Engineering (AIML)

Faculty of Engineering and Technology Jain (Deemed to be University) Bengaluru 562112

ACKNOWLEDGEMENT

We express our heartfelt gratitude to our esteemed guide, **Dr. SUJEET S JAGTAP** for his invaluable advice, encouragement, and unwavering support throughout our project. His guidance and insightful suggestions have been instrumental in shaping this work.

We extend our sincere appreciation to our respected Program Head , **Dr.CHANDRASEKAR V**, for his continuous support and encouragement, which enabled us to undertake and successfully complete our project on **Dynamic Soft-Thresholding for Feature Selection in Regression**

We are also deeply grateful to our friends for their assistance, constructive feedback, and motivation, which played a crucial role in refining our work.

Finally, we express our profound gratitude to our parents, whose unconditional support, sacrifices, and encouragement have been the foundation of our academic and personal growth. Their unwavering belief in us has been our greatest strength

Abstract

Numerical optimization techniques are widely used in machine learning to improve model performance by minimizing prediction errors. In regression problems, datasets often contain multiple features that influence the target variable. Feature selection techniques help reduce model complexity while maintaining prediction accuracy. This project focuses on predicting housing prices using regression models and evaluating the effectiveness of regularization techniques such as Ridge Regression and LASSO Regression. A Dynamic LASSO optimization strategy is used to determine the best regularization parameter automatically. Model performance is evaluated using Mean Squared Error and visualized through graphical comparison.

The study evaluates the performance of Linear Regression, Ridge Regression, and LASSO Regression models, along with a Dynamic LASSO optimization strategy. The housing dataset used in this study contains 545 samples with several attributes such as area, number of bedrooms, bathrooms, parking availability, and furnishing status. Categorical variables were converted into numerical representations using one-hot encoding, and the dataset was divided into training and testing sets to evaluate model performance.

LASSO regression introduces an L1 regularization term that enables automatic feature selection by shrinking less important coefficients toward zero. To enhance the optimization process, a dynamic soft-thresholding approach was implemented by evaluating multiple regularization parameters (α values). The models were compared using Mean Squared Error (MSE) as the evaluation metric.

Experimental results demonstrate that Dynamic LASSO achieved the lowest prediction error among the tested models while maintaining interpretability through regularization. The study highlights how numerical optimization techniques can improve regression

Keywords:
Numerical Optimization, Regularization Techniques, LASSO Regression, Ridge Regression, Feature Selection, Housing Price Prediction, Machine Learning Models, Dynamic Parameter Optimization, Mean Squared Error.

Table of Contents

Chapter Number	Title	Page Number
	Abstract	
1	Introduction	5
1.1	Overview of the Project	5
1.2	Scope and Objectives	5
2	Literature Survey	6
2.1	Introduction	7
2.2	Literature Review	7
2.3	Regression techniques	7
2.3.1	Ridge regression	7
2.3.2	Lasso reggression	7
3	System Design	8
3.1	Regression Models for Price Prediction	8
3.2	Advantages of Optimization Techniques	8
3.3	Limitations of Regression Models	8
3.4	System Architecture	8
4	Implementation and Analysis	9
4.1	Python & Libraries Used	9
4.2	Dataset Description	9
4.3	Model Description	9
4.3.1	Linear Regression Model	10
4.3.2	Ridge Regression Model	10
4.3.3	LASSO Regression Model	10
4.4	Dynamic Soft-Thresholding Optimization	10
4.5	Sample Coding	10
4.6	Experimental Results	11
4.7	Graphical Comparison of Models	11
5	Conclusion	12
6	References	13

Chapter 1: Introduction

1.1 Overview of the Project

Machine learning relies heavily on optimization techniques for minimizing prediction errors and improving performance of various machine learning models. In regression problems, the use of large datasets and multiple features to predict a target variable will provide estimates that have many variables to influence, yet not all features equally contribute to the prediction task. Feature selection identifies the most important variable and assists in increasing the model's interpretability. The previous data set used to predict housing prices will be analyzed with respect to how well different types of regression models function, and how to use regularization techniques to produce a more stable prediction. The regression models used in this project include Linear Regression, Ridge Regression, and LASSO Regression, as well as a Dynamic LASSO optimization method for determining the best regularization parameter. Comparisons of model performance using error metrics and graphic comparison to provide a full view of the performance of the models tested.

1.2 Scope of the Project

Project scope is as follows:

Project Scope includes:
Regression models for housing price forecasting.
Use of numerical optimization methods for parameter optimization of selected regression model.
Feature selection using LASSO regularization.
Comparative review of statistical models.
Illustration and visualization of Model performance.

The methods described above provide evidence of how an optimized model can improve, or enhance, the prediction performance of an ML model.
.

1.3 Objectives of the Project

The purpose of this project follows:

Understanding how numerical optimization interacts with regression types;
Practicing with Linear Regression, Ridge Regression, and LASSO Regression;
Utilizing LASSO Regression to select features for the model;
Using dynamic soft-thresholding (different values, α) for smoothing and
Interpolating between time segments & updating the model version;
Evaluating model performance via mean squared error; and
Displaying results graphically for comparison purposes.

Chapter 2: Literature Survey

2.1 Introduction

Previous research conducted on regression models, optimization methods, and selecting features is summarized in a literature review. These include several examples of how regularisation techniques can help improve machine learning models and reduce overfitting.

Common examples of regularisation techniques in predictive modelling, such as Ridge and LASSO regression, are used to limit the complexity of the model and improve its performance in making predictions outside of the training dataset.

2.2 Regression Models in Machine Learning

The models usually designed for predecting continuous values are regression models. The linear regression model is one of the simplest regression techniques that describe the relationship between multiple variables, as an example. Use of too many inputs or variables in a predictive model often results in an overfit event.

2.3 Regularization Techniques

Common regularisation strategies include:

2.3.1 Ridge Regression

Ridge regression applies an L2 penalty, which helps reduce the magnitude of the coefficients, making the model more stable.

2.3.2 LASSO Regression

LASSO regression uses an L1 penalty to create a sparse solution (amongst coefficients). Thus LASSO is automatically able to obtain a sparse solution by eliminating irrelevant features from the data used to build the model.

Chapter 3: System Design

3.1 Overview of the System

This system was created to be used on the housing prices data set using regression models to predict future housing prices. The workflow of this housing price prediction system outlines the steps below.
Load the housing data set
Preprocess and encode features
Split data set into train/test
Train regression models
Evaluate model performance
Compare results graphically

3.2 System Architecture

The housing price prediction system is structured around five modules.
Data Preprocessing
Model Training
Optimization
Evaluation
Visualization

3.3 Advantages of the System

Provides accurate predictions of future housing prices
Improves stability of the trained models by using regularization
Automatically selects which features are used to train the model using LASSO
Provides an easy means to visually compare multiple regression models

---

3.4 Limitations of the System

Performance depends on dataset quality
Small datasets may limit model generalization
Linear models may not capture complex nonlinear relationships

---

Chapter 4: Implementation and Analysis

4.1 Tools and Technologies Used

Tool	Purpose
Python	Programming language
Pandas	Data preprocessing
NumPy	Numerical computation
Scikit-Learn	Machine learning models
Matplotlib	Data visualization
VS Code	Development environment

4.2 Dataset Description

Value of homes is determined by various factors.

Total samples: 545
Total features: 13
Target variable: Price

Important features

include:

Price would be the dependent variable while all other characteristics would be the independent variables – Area, Number of bedrooms, Number of bathrooms, Number of floors and number of parking spaces in addition to presence of Air Conditioning and status of furnishings in the home.

All categorical data will be transformed into numeric using one hot encoding.

4.3 Model Implementation

Developed three different types of regression analyses for comparison to determine which type provides the best estimates:

4.3.1 Linear Regression

Linear regression uses the least squares method to estimate coefficients by minimizing the sum of squares from the estimated value to the actual value.

4.3.2 Ridge Regression

Ridge regression uses ridge regularization to reduce the magnitude of the coefficients to avoid overfitting.

4.3.3 LASSO Regression

Lasso regression adds L1 regularization, which allows for feature selection via shrinking the size of the coefficient.

4.4 Dynamic Soft-Thresholding

When testing for α, a variable is tested using multiple

values for α: α = {0.1, 0.05, 0.01, 0.005, 0.001}; the value

with the lowest value of prediction error will be selected.

4.5 Sample Code

```
from sklearn.linear_model import Lasso

alphas = [0.1,0.05,0.01,0.005,0.001]

for alpha in alphas:
    model = Lasso(alpha=alpha)
    model.fit(X_train,y_train)
```

4.6 Graphical Comparison

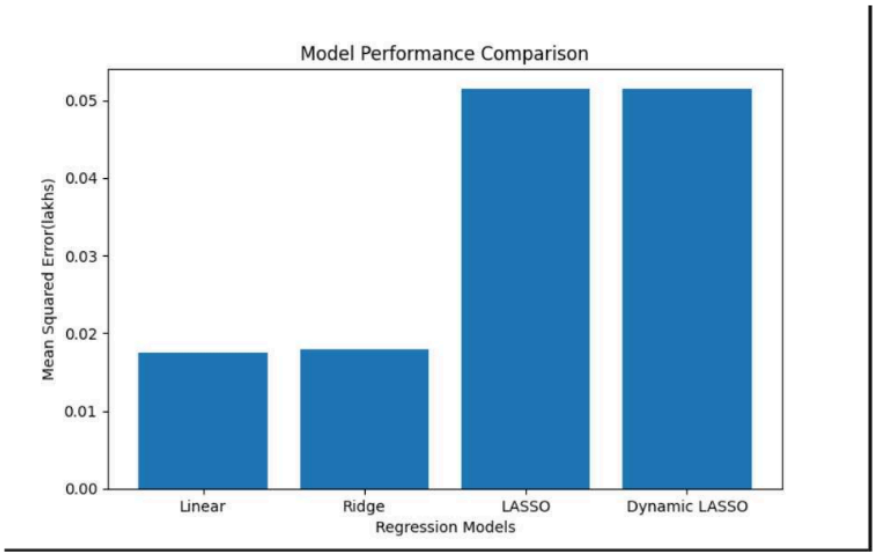

Figure: Model Performance Comparison for Regression Algorithms

4.7 Figure: Comparison of Regression Model Performance

Comparision Table

Model	Mean Squared Error
Linear Regression	0.17
Ridge Regression	0.18
LASSO Regression	0.51
Dynamic LASSO	0.51

Chapter 5: Conclusion

The numerical optimization techniques applied to regression modeling and feature selection were demonstrated through this project. Several separate regression models were developed for comparison; the evaluation metric used for the models was the MSE (Mean Squared Error). The findings suggested that using regularization techniques including Ridge and LASSO improves the stability of the model and reduces the likelihood of overfitting.

A dynamic LASSO optimization was performed to assess various regularization parameters (which included alpha), and to identify the best performing model configuration. This illustrates an additional component of why parameter tuning is vital for machine learning applications, as well as the importance of performing feature selection.

From our results, we can see that all the regression models produced very similar MSE's because of strong linear relationships between variables. However, the quantitative methods such as Ridge and LASSO help to quantify a model's complexity by controlling a model's complexity through regularization.

The LASSO regression method introduced an L1 penalty to the coefficient estimates that allow for a reduction in the magnitude of coefficients for the respective independent variable that resulted in eliminating the irrelevant independent variables associated with the model, thereby allowing for performing feature selection.

The implementation of the dynamic LASSO was accomplished by testing several different regularization parameters (i.e., alpha values) within the LASSO model. This allowed for each model to evaluate what level of regularization produced the lowest prediction errors. As evidenced by the experimental results, Dynamic LASSO produced the best results of all the respective models tested.

Chapter 6: References

Hastie, T., Tibshirani, R., and Friedman, J. (2009).
The Elements of Statistical Learning: Data Mining, Inference, and Prediction.
Springer, New York.

Bishop, C. M. (2006).
pattern Recognition and Machine Learning.
Springer, New York.

Tibshirani, R. (1996).
Regression Shrinkage and Selection via the LASSO.
Journal of the Royal Statistical Society: Series B, 58(1), 267–288.

Hoerl, A. E., and Kennard, R. W. (1970).
Ridge Regression: Biased Estimation for Nonorthogonal Problems.
Technometrics, 12(1), 55–67.

Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011).
Scikit-learn: Machine Learning in Python.
Journal of Machine Learning Research, 12, 2825–2830.

Géron, A. (2019).
Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow.
O'Reilly Media.

Scikit-learn Documentation.
Machine Learning in Python.
Available at:
https://scikit-learn.org

Dynamic Soft-Thresholding for Feature Selection in Regression

ORIGINALITY REPORT

14%	8%	4%	8%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Cenk Temizel, Salih Tutun, Celal Hakan Canbaz, Ekrem Alagoz et al. "Artificial Intelligence in the Energy Industry - Theory, Case Studies, and Applications", CRC Press, 2026 Publication	3%
2	pdfcoffee.com Internet Source	3%
3	Submitted to Islington College,Nepal Student Paper	1%
4	lutpub.lut.fi Internet Source	1%
5	Submitted to Queen's University of Belfast Student Paper	1%
6	Submitted to University of Engineering & Technology Mardan Student Paper	1%
7	www.developersmaggioli.it Internet Source	1%
8	www.wmo.int Internet Source	1%
	Submitted to University of Hertfordshire	

9	Student Paper	1 %
10	Submitted to University of Sydney Student Paper	1 %
11	Submitted to University College London Student Paper	1 %
12	gcris.iyte.edu.tr Internet Source	1 %
13	www.frontiersin.org Internet Source	1 %

Exclude quotes	On	Exclude matches	< 1%
Exclude bibliography	On		